



**SYDNEY BOYS HIGH
SCHOOL**
MOORE PARK, SURRY HILLS

2007

**YEAR 12 Mathematics Extension 1
HSC Task #3**

Mathematics Extension 1

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Each Section is to be returned in a separate bundle.
- Marks may **NOT** be awarded for messy or badly arranged work.
- All necessary working should be shown in every question.

Total Marks – 70

- Attempt questions 1-3
- Start each new section of a separate writing booklet

Examiner: *D.McQuillan*

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

QUESTION 1 (23 marks)**Marks**

(a) Evaluate $\cos^{-1}\left(\frac{2}{5}\right)$ in radians to 4 significant figures.

1

(b) Use the table of integrals to find

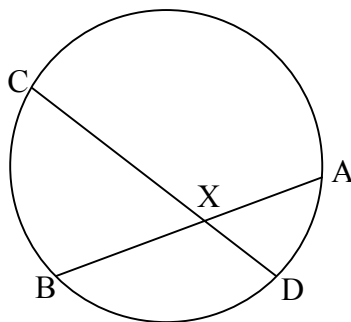
(i) $\int \frac{dx}{9+x^2}$

1

(ii) $\frac{d}{dx}\left(\sin^{-1}\frac{x}{7}\right)$

1

(c)



If $AX = 3$, $BX = 4$ and $CX = 6$ find DX .

1

(d) Find the inverse functions of the following

(i) $f(x) = \frac{x+1}{4}$

1

(ii) $g(x) = \frac{x+2}{1-x}$

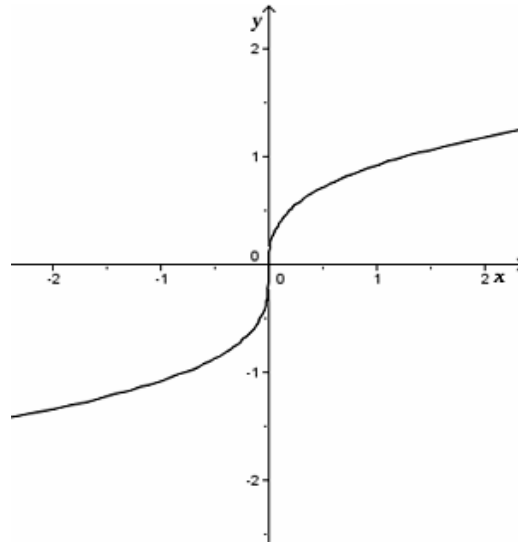
1

(e) Evaluate $\sin^{-1}\left(\sin\frac{2\pi}{3}\right)$.

1

- (f) It is obvious that, $\sqrt{4} < \sqrt{7} < \sqrt{9}$. Use two application of the ‘halving the interval’ method on the equation $x^2 - 7 = 0$ to find a smaller interval containing $\sqrt{7}$. 2

(g)



The diagram shows the graph of $y = \sqrt[3]{x}$. Copy the diagram into your examination booklet. On your diagram sketch the graph of the inverse function of $y = \sqrt[3]{x}$. 2

- (h) Find $\int 8x(x^2 + 2)^3 dx$ using the substitution $u = x^2 + 2$. 2

- (i) A rock is thrown into a still pond and causes a circular ripple. If the radius of the ripple is increasing a 0.5 m/s.

- (i) Find the rate of change of the area as a function of the radius. 2
- (ii) How fast is the area increasing when the radius is 10 m? 1

(j)

- (i) State the domain and range of $y = 5 \cos^{-1} x$. 2
- (ii) Hence sketch $y = 5 \cos^{-1} x$. 1

(k) Evaluate $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 2 \sin x \cos x \, dx$. 2

(l) Show that $\cos\{\tan^{-1}[\sin(\cot^{-1} x)]\} = \sqrt{\frac{x^2 + 1}{x^2 + 2}}$. 2

End of Question 1

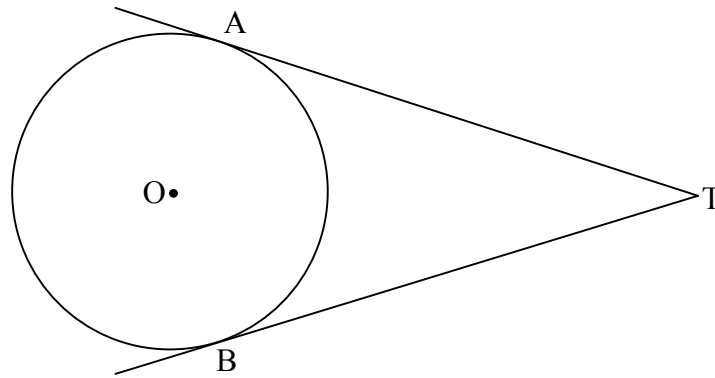
QUESTION 2 (23 marks)

Marks

- (a) Evaluate $\lim_{x \rightarrow \infty} \tan^{-1}(x - x^2)$

1

- (b)



TA and TB are tangents to a circle centre O.

- (i) Prove that ATBO is a cyclic quadrilateral.
- (ii) Hence or otherwise prove that $\angle ATB = 2\angle ABO$.

2

2

- (c) It has been proposed that for all positive integer values of n the following statement is true.

$$2 + 4 + 6 + \dots + 2n = (n + 2)(n - 1)$$

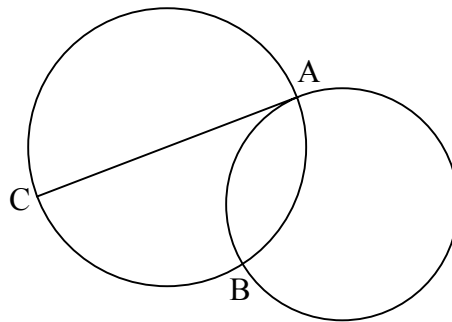
- (i) Assume that the statement is true for $n = k$ and show that it true for $n = k + 1$.
- (ii) Is the original statement true for all positive integer values of n ? Explain your answer.

2

1

- (d) A spot of light, undergoing simple harmonic motion on a computer screen, has a period of π seconds. When the spot is $3\sqrt{3}$ centimetres to the left of its equilibrium point it has a velocity of 6 centimetres per second towards its equilibrium point.
- (i) Show that $x(t) = A \sin(nt) + B \cos(nt)$ is a solution of the simple harmonic motion equation $\ddot{x}(t) = -n^2 x(t)$. 2
 - (ii) Using the initial conditions of the spot of light find values for n , A and B to show that the function $x(t) = 3 \sin(2t) - 3\sqrt{3} \cos(2t)$ describes the motion of the spot. 3
 - (iii) At what time will the spot first reach it's equilibrium point? 2
 - (iv) At what time will the spot first reach it's maximum amplitude? 2
 - (v) What is the maximum amplitude of the spot of light? 1

(e)



The diagram shows two circles intersecting at A and B. The diameter of one circle is AC. Copy the diagram into your examination booklet.

- (i) On your diagram draw a straight line through A, parallel to CB, to meet the second circle at D. 1
- (ii) Prove that BD is a diameter of the second circle. 2
- (iii) Suppose that BD is parallel to CA. Prove that the circles have equal radii. 2

End of Question 2

Start a new examination booklet

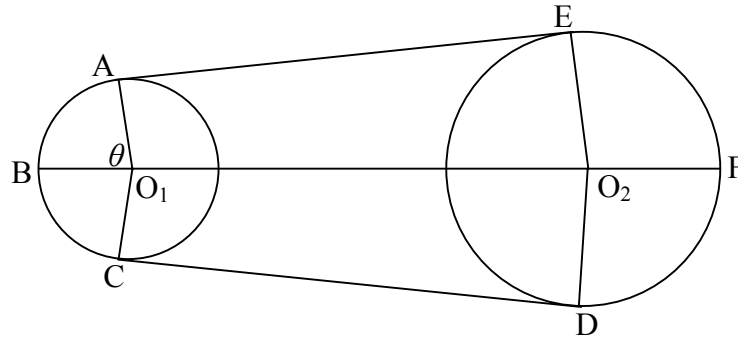
QUESTION 3 (22 marks)

Marks

- (a) Evaluate $4 \int_0^r \sqrt{r^2 - x^2} dx$ using the substitution $x = r \cos \theta$.

3

(b)



A belt $ABCDEF$ of length 40 m passes round two pulleys, centres O_1 and O_2 . The parts ABC and DEF of the belt are in contact with the pulleys and the parts AE and CD are straight.

- (i) If $O_1A = 2$ m, $O_2D = 5$ m, and $\angle AOB = \theta$ radians, show that

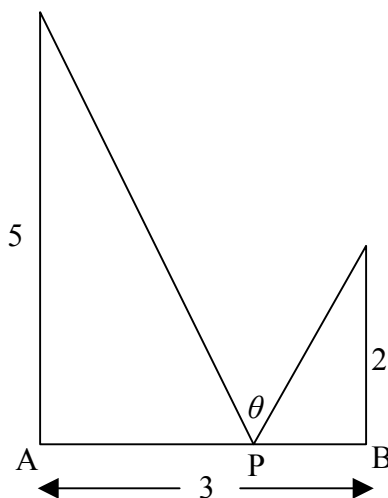
3

$$3 \tan \theta = 3\theta + 20 - 5\pi.$$

- (ii) Using a first approximation of $\frac{\pi}{3}$ and one application of Newton's method, find a root of the equation to 2 decimal places.

2

- (c) A particle is moving such that its velocity, v m/s, is related to its displacement, x metres, by the formula $v = 2x - 3$.
- (i) Show that the acceleration, a m/s², of the particle is given by $a = 4x - 3$. 2
- (ii) Initially the particle has displacement $x = 3.5$ m. Show that the relationship between displacement, x , and time, t seconds, can be expressed as
- $$\log_e \left(\frac{2x - 3}{4} \right) = 2t.$$
- 3
- (iii) Write down the function for acceleration in terms of t . 1
- (d) Use mathematical induction to show that $n! \geq 2^{n-1}$ for $n = 1, 2, 3, \dots$ 4
- (e) How far from A should the point P be, on the interval AB, so as to maximise the angle θ . 4



End of Question 3

End of Exam

2007 Mathematics Extension 1 Assessment 3: Solutions Question 1

1. (a) Evaluate $\cos^{-1}\left(\frac{2}{5}\right)$ in radians to 4 significant figures.

1

Solution: $1.159279480727408599846583794\dots \approx 1.159$.

- (b) Use the table of integrals to find

(i) $\int \frac{dx}{9+x^2}$

1

Solution: $\int \frac{dx}{9+x^2} = \frac{1}{3} \tan^{-1} \frac{x}{3} + c.$

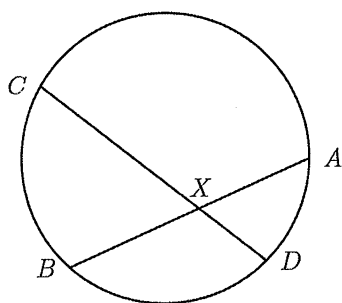
(ii) $\frac{d}{dx} \left(\sin^{-1} \frac{x}{7} \right)$

1

Solution: $\frac{d}{dx} \left(\sin^{-1} \frac{x}{7} \right) = \frac{1}{\sqrt{49-x^2}}.$

- (c)

1



If $AX = 3$, $BX = 4$ and $CX = 6$, find DX .

Solution: $DX \times XC = AX \times XB$ (product of intercepts theorem).
 $DX \times 6 = 3 \times 4,$
 $\therefore DX = 2.$

(d) Find the inverse functions of the following

(i) $f(x) = \frac{x+1}{4}$

1

Solution:

$$x = \frac{f^{-1}(x) + 1}{4},$$
$$4x = f^{-1}(x) + 1,$$
$$f^{-1}(x) = 4x - 1.$$

(ii) $g(x) = \frac{x+2}{1-x}$

2

Solution:

$$x = \frac{g^{-1}(x) + 2}{1 - g^{-1}(x)},$$
$$x(1 - g^{-1}(x)) = g^{-1}(x) + 2,$$
$$x - 2 = g^{-1}(x) + xg^{-1}(x),$$
$$g^{-1}(x) = \frac{x-2}{x+1}.$$

(e) Evaluate $\sin^{-1}\left(\sin \frac{2\pi}{3}\right)$.

1

Solution: $\sin \frac{2\pi}{3} = \sin \frac{\pi}{3}$ and the range of inverse sine is from $-\frac{\pi}{2}$ to $\frac{\pi}{2}$,
 $\therefore \sin^{-1}\left(\sin \frac{2\pi}{3}\right) = \frac{\pi}{3}.$

(f) It is obvious that $\sqrt{4} < \sqrt{7} < \sqrt{9}$. Use two applications of the 'halving the interval' method on the equation $x^2 - 7 = 0$ to find a smaller interval containing $\sqrt{7}$.

2

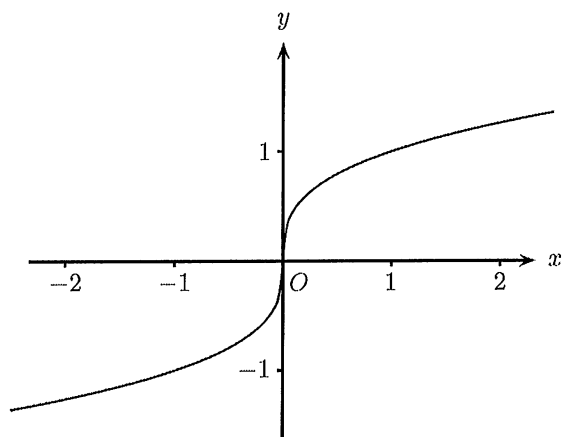
Solution: Put $f(x) = x^2 - 7$,

$$f(2) = -3,$$
$$f(3) = 2,$$
$$f(2.5) = -0.75,$$
$$(3 + 2.5)/2 = 2.75,$$
$$f(2.75) = 0.5625.$$

$\therefore$ a better estimate is between 2.5 and 2.75.

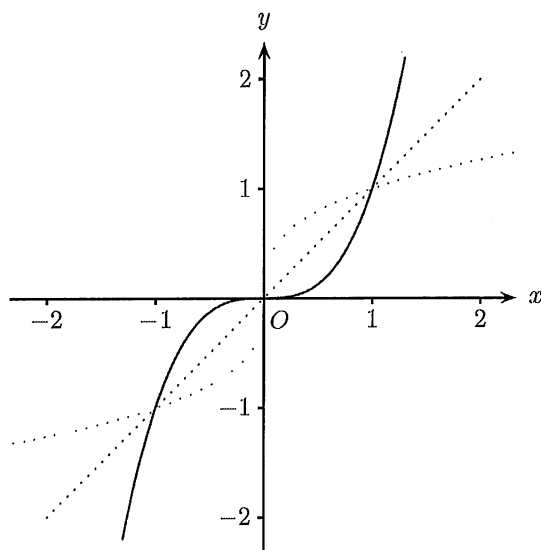
(g)

2



The diagram shows the graph of $y = \sqrt[3]{x}$. Copy the diagram into your writing booklet. On your diagram sketch the graph of the inverse function of $y = \sqrt[3]{x}$.

Solution:



(h) Find $\int 8x(x^2 + 2)^3 dx$ using the substitution $u = x^2 + 2$.

2

Solution:	$\int 8x(x^2 + 2)^3 dx = \int 4u^3 du,$	$u = x^2 + 2$
	$= u^4 + c,$	$\frac{du}{dx} = 2x$
	$= (x^2 + 2)^4 + c.$	

- (i) A rock is thrown into a still pool and causes a circular ripple. The radius of the ripple is increasing at 0.5 m/s.

- (i) Find the rate of change of the area as a function of the radius.

2

Solution: Area, $A = \pi r^2$,
 $\frac{dA}{dr} = 2\pi r$.
 $\frac{dr}{dt} = 0.5 \text{ m/s (given)}$
 $\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$,
 $= 2\pi r \times 0.5$,
 $= \pi r$.

- (ii) How fast is the area increasing when the radius is 10 m?

1

Solution: The area is increasing at $10\pi \text{ m}^2/\text{s}$, or about $31.4 \text{ m}^2/\text{s}$.

- (j) (i) State the domain and range of $y = 5 \cos^{-1} \frac{x}{3}$.

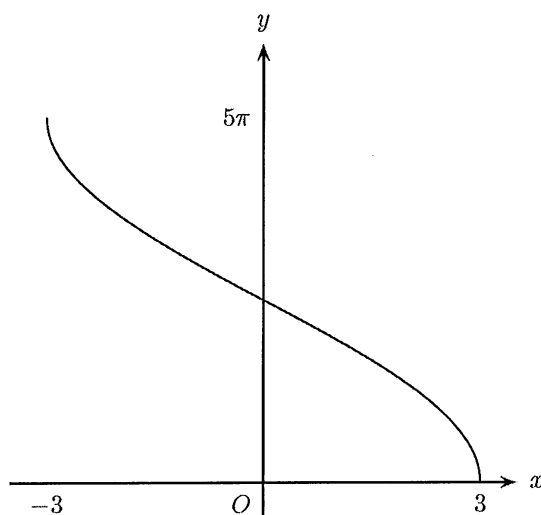
2

Solution: $-1 \leq \frac{x}{3} \leq 1$,
 $-3 \leq x \leq 3$.
 $0 \leq y \leq 5\pi$.

- (ii) Hence sketch $y = 5 \cos^{-1} \frac{x}{3}$.

1

Solution:



(k) Evaluate $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 2 \sin x \cos x \, dx$.

2

Solution: Method 1:

$$\begin{aligned} \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 2 \sin x \cos x \, dx &= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin 2x \, dx, \\ &= \left[\frac{-\cos 2x}{2} \right]_{\frac{\pi}{4}}^{\frac{\pi}{2}} \\ &= -\frac{1}{2} \{-1 - 0\}, \\ &= \frac{1}{2}. \end{aligned}$$

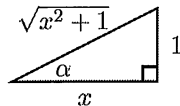
Solution: Method 2:

$$\begin{aligned} \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 2 \sin x \cos x \, dx &= \int_{\frac{1}{\sqrt{2}}}^1 2u \frac{du}{dx} \, dx, & \text{Put } u &= \sin x, \\ &= [u^2]_{\frac{1}{\sqrt{2}}}^1, & \frac{du}{dx} &= \cos x. \\ &= 1 - \frac{1}{2}, & \text{When } x &= \frac{\pi}{2} \quad u = 1, \\ &= \frac{1}{2}. & x &= \frac{\pi}{4} \quad u = \frac{1}{\sqrt{2}}. \end{aligned}$$

(l) Show that $\cos \{ \tan^{-1} [\sin (\cot^{-1} x)] \} = \sqrt{\frac{x^2 + 1}{x^2 + 2}}$.

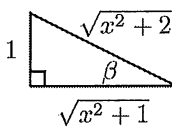
2

Solution: We note that $x \in \mathbb{R}$ and $\cot^{-1} x$ ranges over the interval $(0, \pi)$. Over this domain $\sin \alpha = \sin \left(\frac{\pi}{2} - \alpha \right)$.



From the diagram, $\sin \alpha = \frac{1}{\sqrt{x^2 + 1}}$, with range $0 < \sin \alpha < 1$.

Over this domain, putting $\beta = \tan^{-1}(\sin \alpha)$, $0 < \beta < \frac{\pi}{4}$.

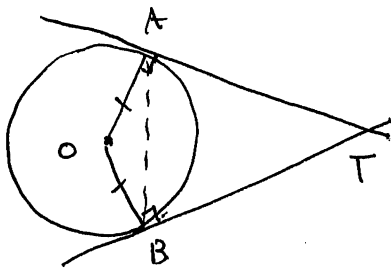


Now $\cos \beta = \frac{\sqrt{x^2 + 1}}{\sqrt{x^2 + 2}}$, i.e. $\cos \{ \tan^{-1} [\sin (\cot^{-1} x)] \} = \sqrt{\frac{x^2 + 1}{x^2 + 2}}$.

Question (2)

$$\begin{aligned} (a) \lim_{x \rightarrow \infty} \tan^{-1}(x-x^2) \\ = \lim_{x \rightarrow \infty} \tan^{-1}(-x^2) \\ = -\frac{\pi}{2} \end{aligned} \quad (1)$$

(b)



$$(i) \angle OAT = \angle OBT = 90^\circ$$

(Tgt perpendicular to the radius at the pt of contact)

$\therefore$ OATB is cyclic

(Opp. $\angle$'s of OATB are supplementary)

- (ii) Join AB, let $\angle OBA = \alpha$
 $\therefore \angle OAB = \alpha$ ($\triangle OAB$ is isosceles $OA = OB$)
 $\therefore \angle AOB = 180 - 2\alpha$ ($\angle$ sum of $\triangle$)
 $\therefore \angle ATB = 2\alpha$ (Opp. $\angle$'s of a cyclic quad. are supplementary.)
 $\therefore \angle ATB = 2\angle ABO$

$$(c) (i) 2 + 4 + \dots + 2k + 2(k+1)$$

$$= (k+2)(k-1) + 2(k+1)$$

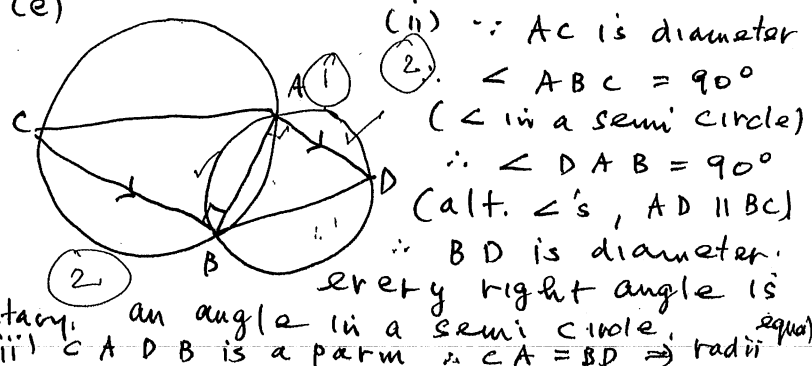
$$= k^2 + 3k = [(k+1)+2][(k+1)-1]$$

$\therefore S(k+1)$ is true assume $S(k)$ is true.

(ii) not true for $n=1$, or 2 .

$$n=1, \text{ L.H.S} = 2, \text{ R.H.S} = (1+2)(1-1) = 0 \neq \text{L.H.S.}$$

(e)



(ii) $\therefore AC$ is diameter

(2) $\angle ABC = 90^\circ$

($\angle$ in a semi circle)

$\therefore \angle DAB = 90^\circ$

(alt. $\angle$'s, $AD \parallel BC$)

$\therefore BD$ is diameter.

every right angle is

an angle in a semi circle.

(iii) $CA \parallel DB$ is a parm $\therefore CA = BD \Rightarrow$ radii equal

(d)

$$(i) x(t) = A \sin(nt) + B \cos(nt)$$

$$\dot{x} = nA \cos(nt) - nB \sin(nt)$$

$$\ddot{x} = -n^2 A \sin(nt) - n^2 B \cos(nt)$$

$$\ddot{x} = -n^2 [A \sin(nt) + B \cos(nt)]$$

$$= -n^2 x(t)$$

$$(ii) T = \frac{2\pi}{n} \quad \pi = \frac{2\pi}{n}$$

$$(x) \therefore n = 2.$$

$$\text{Let } x(t) = R \sin(2t + \alpha)$$

expanding:

$$= (R \cos \alpha) \sin 2t + (R \sin \alpha) \cos 2t$$

$$\therefore A = R \cos \alpha = 3$$

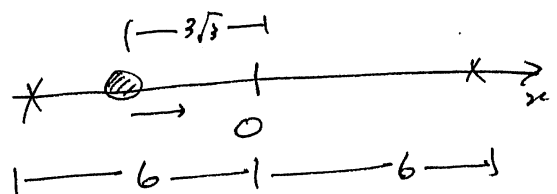
$$B = R \sin \alpha = -3\sqrt{3}$$

$$\Rightarrow \tan \alpha = -\sqrt{3} \Rightarrow \alpha = -\frac{\pi}{3}$$

$$\therefore R = 6$$

$$x(t) = 6 \sin(2t - \frac{\pi}{3})$$

$$\dot{x}(t) = 12 \cos(2t - \frac{\pi}{3})$$



$$(b) \text{ When } x(t) = 0$$

$$0 = \sin(2t - \frac{\pi}{3})$$

$$\therefore 2t - \frac{\pi}{3} = 0, \pi, 2\pi, \dots$$

$$2t = \frac{\pi}{6} \Rightarrow t = \frac{\pi}{6}$$

$$(c) \text{ When } x(t) = 6$$

$$6 = 6 \sin(2t - \frac{\pi}{3})$$

$$\therefore \sin(2t - \frac{\pi}{3}) = 1$$

$$\therefore 2t - \frac{\pi}{3} = \frac{\pi}{2}$$

$$\therefore 2t = \frac{5\pi}{6} \Rightarrow t = \frac{5\pi}{12}$$

$$(d) R = 6$$

QUESTION 3

$$\begin{aligned} 3(a) \quad x &= r \cos \theta & \frac{dx}{d\theta} &= -r \sin \theta \\ 4 \int_0^r \sqrt{r^2 - x^2} \, dx & & x=0 \quad \theta &= \frac{\pi}{2} \\ & & x=r \quad \theta &= 0 \\ &= 4 \int_{\frac{\pi}{2}}^0 \sqrt{r^2 - r^2 \cos^2 \theta} \cdot -r \sin \theta \, d\theta \\ &= 4 \int_0^{\frac{\pi}{2}} r \sin \theta \cdot r \sin \theta \, d\theta \\ &= \int_0^{\frac{\pi}{2}} 4r^2 \sin^2 \theta \, d\theta = 4r^2 \int_0^{\frac{\pi}{2}} \frac{1 - \cos 2\theta}{2} \, d\theta \\ &= 4r^2 \left[\frac{1}{2} \theta - \frac{1}{4} \sin 2\theta \right]_0^{\frac{\pi}{2}} \\ &= 4r^2 \left[\frac{\pi}{4} - 0 - (0) \right] = \pi r^2 \end{aligned}$$

(b) (ii) $V = 2x - 3$ $\frac{1}{2}V^2 = \frac{1}{2}(4x^2 - 12x + 9)$
 $\frac{d(\frac{1}{2}V^2)}{dx} = a = 4x - 6$

ii)

$$v = \frac{dx}{dt} = 2x - 3$$
$$\frac{dt}{dx} = \frac{1}{2x-3}$$
$$t = \frac{1}{2} \log_e(2x-3) + C$$
$$t=0 \quad x=3.5 \quad C = -\frac{1}{2} \log 4$$
$$t = \frac{1}{2} \log_e \left(\frac{2x-3}{4} \right)$$
$$2t = \log_e \left(\frac{2x-3}{4} \right)$$
$$\frac{2x-3}{4} = e^{2t}$$
$$2x-3 = 4e^{2t}$$

$$\begin{aligned} \text{(iii)} \quad a &= 4x - 6 \\ &= 2 \times 4e^{2t} \\ &= 8e^{2t} \end{aligned}$$

(c) Assume $P(k)$ is true

(i) $k! \geq 2^{k-1}$ for $k = 1, 2, 3, \dots$

$$1! \geq 2^0 \therefore P(1) \text{ is true}$$

if $n = k+1$:

$$(k+1)! = (k+1) \cdot k!$$

$$\geq (k+1) 2^{k-1} \text{ using assumption}$$

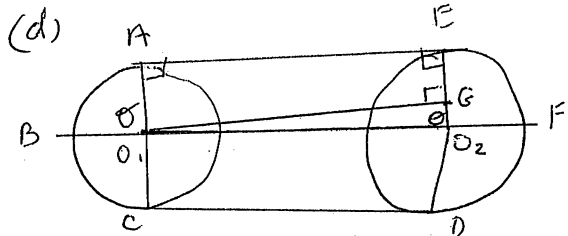
$$\geq 2 \cdot 2^{k-1} \text{ as } k \geq 1$$

$$(k+1)! > 2^k$$

$\therefore$ If $P(k)$ is true, $P(k+1)$ is true.

by Mathematical Induction

$$n! \geq 2^{n-1} \text{ for } n = 1, 2, 3, \dots$$


$$BO_2E = \Theta \text{ (cointerior to } AO, O_2)$$

Draw $O, G \parallel AE$

$$\tan \theta = \frac{O_1G}{O_2G} = \frac{AE}{5-2} = \frac{AE}{3}$$

$$AE = 3 \tan \theta$$

$$I = r\theta : \text{Arc } ABC = 2 \times 20 = 40$$

$$A_{\text{REFD}} = 5[2(\pi - \theta)]$$

$$= 10\pi - 10\theta$$

$$\text{Belt} = \pi r ABC + 2AE + \pi r CDEF = 40\text{m}$$

$$4\theta + 6\tan\theta + 10\pi - 10\theta = 40$$

$$6 + \arg \theta = 6\theta + 40 - 10\pi$$

$$3 \tan \theta = 3\theta + 20 - 5\pi$$

$$(d) \quad a_1 = a - \frac{f(\theta)}{f'(\theta)}$$

(ii)

$$f(\theta) = 3 \tan \theta - 3\theta - 20 + 5\pi$$

$$f'(\theta) = 3 \sec^2 \theta - 3$$

$$a_1 = \frac{\pi}{3} - \frac{f(\frac{\pi}{3})}{f'(\frac{\pi}{3})}$$

$$= \frac{\pi}{3} - \left(\frac{-2.24}{9} \right)$$

$$= 1.295 \text{ (3 d.p.)}$$

$$(e) \quad \theta = \pi - \tan^{-1}\left(\frac{5}{x}\right) - \tan^{-1}\left(\frac{2}{3-x}\right)$$

$$\frac{d\theta}{dx} = -\frac{1}{1+(\frac{5}{x})^2} \cdot \frac{5}{x^2} - \frac{2}{1+(\frac{2}{3-x})^2} \cdot \frac{1}{(3-x)^2} \cdot (-1)$$

$$= \frac{5}{x^2+25} - \frac{2}{(3-x)^2+4}$$

$$\frac{d\theta}{dx} = 0 \text{ when } 5[(3-x)^2+4] = 2x^2+50$$

$$65 - 30x^2 + 5x^2 = 2x^2 + 50$$

$$3x^2 - 30x + 15 = 0$$

$$x^2 - 10x + 5 = 0$$

$$x = \frac{10 \pm \sqrt{80}}{2}$$

$$= 5 \pm 2\sqrt{5}$$

$$x = 5 - 2\sqrt{5} \quad (x < 3)$$

$$\text{when } x = 0.5 \quad \frac{d\theta}{dx} > 0$$

$$x = 0.6 \quad \frac{d\theta}{dx} < 0$$

$$\therefore \text{max when } x = 5 - 2\sqrt{5}$$
